

# PHYSICAL AI: WHOLE-CURRICULUM ARCHITECTURE ROADMAP

A Principled Journey from Physical Constraints to Frontier Dual-Brain Embodied Systems

## PART I: FOUNDATIONS

### Physics, Silicon Hierarchy

#### Ch 1: The Boundary

The physical AI imperative  
· Causal loop

#### Ch 2: Constraints

5 Physical Columns  
· Latency fresh wall

#### Ch 3: Cognition

5 Cognitive Stages  
· The Great Systems Tension

#### Ch 4: Hierarchy

Multi-rate asynchronous topology  
· 3 Cadences

## PART II: THE 5 ORGANS

### Sensory-Motor Execution Pipeline

#### Ch 5: Perception

Sensory transduction  
· Spatial tokenization

#### Ch 6: Memory

SE(3) World models  
· Dynamic frame trees

#### Ch 7: Reasoning

VLM Grounding  
· Expiring intent leases

#### Ch 8: Planning

Action chunking  
· C<sup>2</sup> jerk continuity

#### Ch 9: Reflexes

Real-time MCU enforcer  
· CBF projection

## PART III: INTEGRATION

### Hardware, Governance Release

#### Ch 10: Placement

4-Tier heterogeneous placement  
· Thermal budgets

#### Ch 11: Governance

Authority arbitration  
· C<sup>2</sup> bumpless takeover

#### Ch 12: Assurance

CAE safety cases  
· 4-Rung qualification ladder

#### Ch 13: Capstone

Dual-brain physical kit  
· Oral defense dossier